Annals of Nuclear Energy 115 (2018) 343–351

Contents lists available at ScienceDirect

Annals of Nuclear Energy

journal homepage: www.elsevier.com/locate/anucene

Magnetic-field analysis of an MHD channel in a liquid-metal circulation system of a prototype GenIV sodium fast reactor

Geun Hyeong Lee, Hee Reyoung Kim *

Ulsan National Institute of Science and Technology, Department of Nuclear Engineering, Ulsan 689-798, Republic of Korea

ARTICLE INFO

Article history:
Received 31 August 2017
Received in revised form 14 December 2017
Accepted 30 January 2018
Available online 22 February 2018

Keywords:
Magnetic flux density
Helical-type DC electromagnetic pump
Permanent magnet
Ferromagnetic material

ABSTRACT

A helical-type direct current (DC) electromagnetic pump with a developed pressure of 10 kPa and flow rate of 0.005 m^3/s was analyzed at a temperature of 226 °C for an active decay heat-removal system (ADHRS) of a prototype GenIV sodium fast reactor. The rectangular-type DC electromagnetic pump, which was selected for the ADHRS, required a very large current (>1000 A). A helical-type DC electromagnetic pump was considered to provide multiple current to a pump duct to reduce the input current requirement. The magnetic flux density of a permanent magnet is one of the important factors that determine the performance of the electromagnetic pump. The magnetic field for the flow channel of the helical-type DC electromagnetic pump was analyzed in which a Sm_2Co_{17} permanent magnet was used to generate the Lorentz force for the circulation of liquid metal. The Lorentz force, which directly affected the developed pressure of the electromagnetic pump, was increased proportional to the magnetic flux density, which led to the increase in the velocity of the liquid metal in the flow channel. The permanent magnet in the +z and −z directions and the ferromagnetic material added in the r direction increased the magnitude of the magnetic flux density, which led the magnetic flux line to the flow channel. The arrangement of the permanent magnet system showed the optimized geometry of the inner ferromagnetic material with a radius of 110 mm and outer ferromagnetic material with a thickness of 30 mm and height of 70 mm. The average value of the magnetic flux density in the liquid-metal flow channel was 0.848 T under maximum condition by considering the mechanical and spatial restrictions.

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1. Introduction

Electromagnetic pumps, instead of mechanical pumps, are used to circulate liquid metal owing to their advantages such as the absence of moving and sealing parts while eliminating the possibility of liquid-metal leak (Kim, 2014). A direct current (DC) electromagnetic pump was considered for an active decay heat-removal system (ADHRS) to transport liquid sodium to a prototype GenIV sodium fast reactor (PGSFR). When normal heat removal from the reactor is not possible, passive decay heat-removal system (PDRC) and ADHRS are used to remove residual heat from the reactor. ADHRS is operated under force convection condition, whereas PDRC is operated under natural convection condition. The ADHRS considers the electromagnetic pump driven by an electromagnetic force generated from the Lorentz force for the forced sodium circulation in the sodium loop, which contains a blower with a finned-tube sodium-to-air heat exchanger (FDHX), as shown in Fig. 1 (Lee, 2013). The helical-type DC electromagnetic pump is adopted to circulate liquid sodium with high electrical conductivity (Ohse, 1985; Baker, 1987) via an electromagnetic force (or Lorentz force) generated from the vector product of the current density and magnetic field (Nashine, 2007; Gutierrez, 1965). This process is adopted because the electromagnetic pump structure has no sealing part and an impeller, which can cause failure during the transportation of high-reactivity liquid sodium. In particular, the helical-type DC electromagnetic pump requires a low current compared with a rectangular-type one because it can obtain multiple current from the geometry of the pump duct. Magnetic flux density is one of the main factors that contribute to the generation of the pumping force from the Lorentz force and the induced electromotive force in the circular channel of the helical-type DC electromagnetic pump, which affects the specification of the flow rate and the developed pressure of the electromagnetic pump (Nashine, 2006; Blake, 1957). The magnetic flux density generated from the combined arrangement of the permanent magnet and ferromagnetic material in the flow channel in the electromagnetic pump is analyzed, where the Lorentz force produced by the vector product of the externally driven current and magnetic field in the liquid-metal flow inside the channel drives the pump (Oka, 2012). The pumping

* Corresponding author.
E-mail address: kimhr@unist.ac.kr (H.R. Kim).

https://doi.org/10.1016/j.anucene.2018.01.049
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Nomenclature

B	Magnetic flux density [T]
B_e	Magnetic flux density of the permanent magnet [T]
$B_{e,r}$	R-direction magnetic flux density of the permanent magnet [A/m^2]
$B_{e,z}$	Z-direction magnetic flux density of the permanent magnet [A/m^2]
B_i	Magnetic flux density of the electrode stub [T]
$B_{i,r}$	R-direction magnetic flux density of the electrode stub [A/m^2]
$B_{i,z}$	Z-direction magnetic flux density of the electrode stub [A/m^2]
$B_{i,\theta}$	θ -direction magnetic flux density of the electrode stub [A/m^2]
B_{n,S_i}	Normal component of magnetic flux density at inner ferromagnetic material region
B_{n,S_m}	Normal component of magnetic flux density at permanent magnet region
B_{n,S_o}	Normal component of magnetic flux density at outer ferromagnetic material region
B_{n,S_p}	Normal component of magnetic flux density at pump duct region
B_t	Total magnetic flux density [T]
E	Electric field [(kg·m)/(s^3·A)]
E_r	R-direction electric field [(kg·m)/(s^3·A)]
E_t	Total electric field [(kg·m)/(s^3·A)]
E_z	Z-direction electric field [(kg·m)/(s^3·A)]
f	Force density [kg/(s^2·m^2)]
H	Magnetic-field intensity [A/m]
H_m	Magnetic-field intensity of the permanent magnet [A/m]
H_t	Tangential component of magnetic-field intensity [A/m]
H_{t,S_i}	Tangential component of magnetic-field intensity at inner ferromagnetic material region
H_{t,S_m}	Tangential component of magnetic-field intensity at permanent magnet region
H_{t,S_o}	Tangential component of magnetic-field intensity at outer ferromagnetic material region
H_{t,S_p}	Tangential component of magnetic-field intensity at pump duct region
J	Current density [A/m^2]
J_r	R-direction current density [A/m^2]
J_t	Total current density [A/m^2]
J_z	Z-direction current density [A/m^2]
J_θ	θ -direction current density [A/m^2]
M	Magnetization [A/m]
p	Developed pressure of pump [A/m]
R_i	Inner ferromagnetic material radius [m]
R_o	Radius up to the outer ferromagnetic material [m]
R_p	Radius up to the pump duct [m]
R_m	Radius up to the permanent magnet [m]
S_i	Inner ferromagnetic material region
S_o	outer ferromagnetic material region
S_p	Pump duct region
S_m	Permanent magnet region
t	Time [s]
v	Velocity of fluid [m/s]
v_θ	θ -direction velocity of fluid [m/s]
ε_0	Permittivity in vacuum [F/m]
μ_0 χ_m	Permeability in vacuum [H/m]
μ_r	Relative permeability
ρ	Density of liquid metal [kg/m^3]
σ	Electrical conductivity of liquid metal [1/(Ω · m)]
χ_m	Magnetic susceptibility

force appears as a function of the magnetic flux density and pump geometrical variables in which a higher magnetic flux density is needed to enhance the hydraulic efficiency of the helical-type DC electromagnetic pump (Kikuchi, 1977). In the present study, the magnetic field is analyzed and optimized in terms of the arrangement of the permanent magnet according to the direction of the magnetic field and the geometrical size of the ferromagnetic material.

2. Analysis of a helical-type DC electromagnetic pump

The helical-type DC electromagnetic pump (Marti, 2015; Lee, 2017) contains four types of components: ferromagnetic material, permanent magnet (Zhu, 1992), electrode stub, and pump duct (Ho, 1977), as shown by the schematic in Figs. 2 and 3. The Lorentz force is generated by the vector product of the current of the electrode stub and the magnetic flux density of the permanent magnet where the liquid metal flows in the θ direction along the helical-type pump flow channel that receives the electromagnetic force generated by the current in the z direction and the magnetic flux density in the r direction, as shown in Fig. 2 (Bennecib, 2009). The Navier-Stokes equation for magnetohydrodynamics (MHD) is used to drive the developed pressure of the electromagnetic pump, as expressed in Eq. (1), and the magnetic flux density used in Eq. (1) is determined by Maxwell's equation of Ampere's law, Faraday's law, and Gauss's law for magnetism and Ohm's law expressed in Eqs. (2)–(5) using the finite element method (FEM) (Moffatt, 1978; Arumugam, 1985; Hughes, 1995).

$$\text{Navier} - \text{Stokes equation}: \rho\left(\frac{\partial}{\partial t} + v \cdot \nabla\right) v = \vec{J}_t \times \vec{B} - \nabla p \tag{1}$$

$$\text{Ampere's law}: \nabla \times \vec{B} = \mu_0\left(\vec{J} + \varepsilon_0 \frac{\partial E}{\partial t}\right) \tag{2}$$

$$\text{Faraday's law}: \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{3}$$

$$\text{Gauss's law for magnetism}: \nabla \cdot \vec{B} = 0 \tag{4}$$

$$\text{Ohm's law}: \vec{J} = \sigma(\vec{E}\theta + \vec{v} \times \vec{B}) \tag{5}$$

The equations of the electric field, current density, magnetic flux density, and velocity of the fluid are expressed in Eqs. (6)–(11). The electric field in the θ direction is eliminated due to the symmetry of the pump duct. The magnetic flux density is divided into external magnetic flux density of the permanent magnet and induced magnetic flux density of the electrode stub. The external magnetic flux density in the θ direction is eliminated because of the symmetry of the permanent magnet and ferromagnetic material. The velocity of the fluid has only a θ-direction component because the pump duct is built in the direction. The z-direction component is negligible because the pump-duct circumference is much higher than the pump-duct radius. In other words, the angle of the pump duct is almost equal to zero.

$$\vec{E}_t(r, \theta, z) = E_r\hat{r} + E_z\hat{z} \tag{6}$$

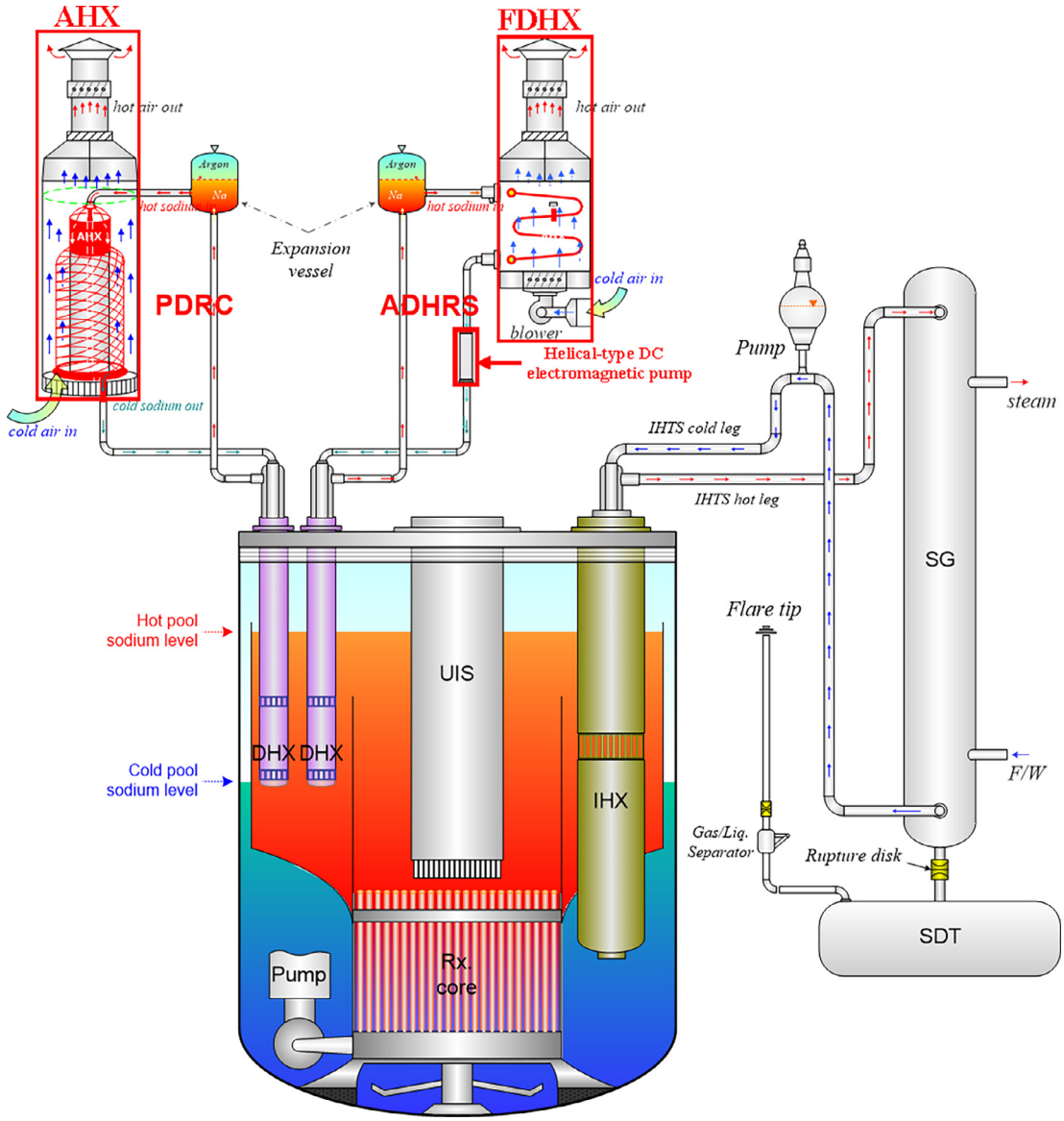

Fig. 1. Schematic of the PGSFR, including the ADHRS.

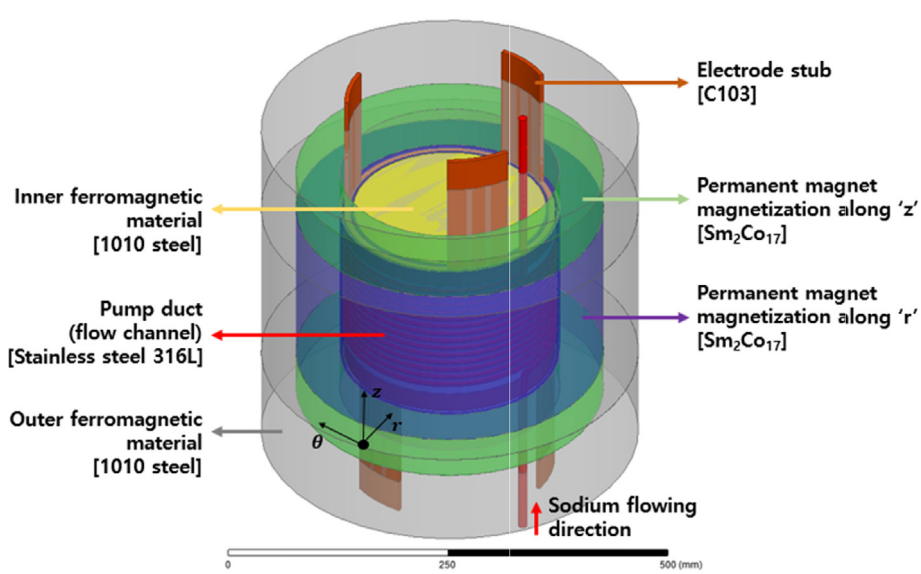

Fig. 2. Schematic of the helical-type DC electromagnetic pump.

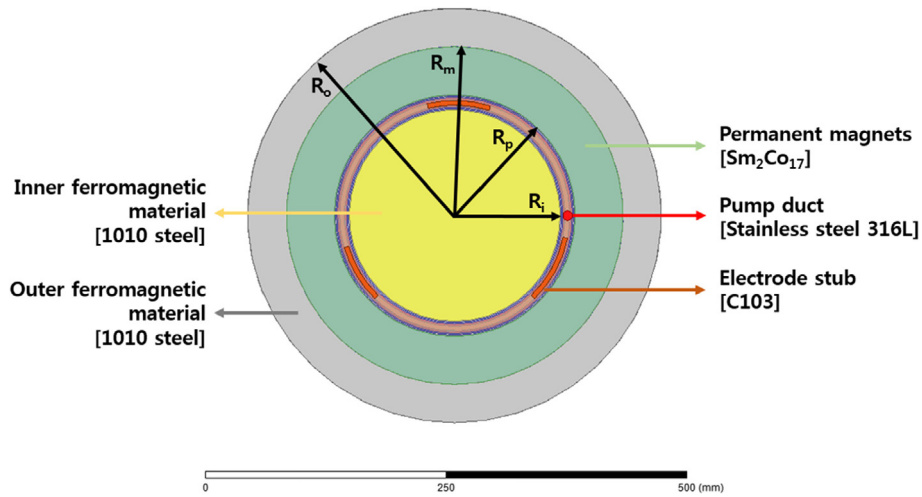

Fig. 3. Schematic of the top-view of the helical-type DC electromagnetic pump.

$$\vec{J}_t(r,\theta,z) = J_r\hat{r} + J_\theta\hat{\theta} + J_z\hat{z} \tag{7}$$

$$\vec{B}_t(r,\theta,z) = \vec{B}_i(r,\theta,z) + \vec{B}_e(r,\theta,z) \tag{8}$$

$$\vec{B}_i(r,\theta,z) = B_{i,r}\hat{r} + B_{i,\theta}\hat{\theta} + B_{i,z}\hat{z} \tag{9}$$

$$\vec{B}_e(r,\theta,z) = B_{e,r}\hat{r} + B_{e,z}\hat{z} \tag{10}$$

$$\vec{v}(r,\theta,z) = v_\theta\hat{\theta} \tag{11}$$

Ampere's law in Eq. (2) is expressed in Eqs. (12)–(15) using the curl operator calculation in a cylindrical coordinate. The term of the magnetic flux density is eliminated because of the symmetry of the permanent magnet, ferromagnetic material, and current density in the flow channel of the helical-type DC electromagnetic pump. The external magnetic flux density is created by some rotating electrons in which the circulating current is microscopically small. Therefore, only the induced magnetic flux density is considered to calculate Ampere's law. The electric field due to the DC remains constant as time changes. Thus, the rate of change in the electric-field term disappears. The magnetic permeability is expressed as

$$\nabla \times \vec{B}_i = \mu_0\vec{J}_t \tag{12}$$

$$\nabla \times \vec{B}_i = \frac{\partial B_{i,\theta}}{\partial z}\hat{r} + \left(\frac{\partial B_{i,r}}{\partial z} - \frac{\partial B_{i,z}}{\partial r}\right)\hat{\theta} + \frac{1}{r}\frac{\partial(rB_{i,\theta})}{\partial r}\hat{z} \tag{13}$$

$$\mu_0\vec{J}_t = \mu_0(J_r\hat{r} + J_\theta\hat{\theta} + J_z\hat{z}) \tag{14}$$

$$\frac{\partial B_{i,\theta}}{\partial z} = \mu_0 J_r, \quad \frac{\partial B_{i,r}}{\partial z} - \frac{\partial B_{i,z}}{\partial r} = \mu_0 J_\theta, \quad \frac{1}{r}\frac{\partial(rB_{i,\theta})}{\partial r} = \mu_0 J_z. \tag{15}$$

In the same manner, Faraday's law in Eq. (3) is expressed by Eqs. (16)–(18). The magnetic flux density is not a time-varying function because the helical-type DC electromagnetic pump employs permanent magnet and DC. The $\frac{\partial}{\partial\theta}$ component of the electric field is also deleted because of the symmetry of the permanent magnet, ferromagnetic material, and current density in the flow channel.

$$\nabla \times \vec{E}_t = 0 \tag{16}$$

$$\nabla \times \vec{E}_t = \left(\frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial r}\right)\hat{\theta} \tag{17}$$

$$\frac{\partial E_z}{\partial r} - \frac{\partial E_r}{\partial z} = 0 \tag{18}$$

Gauss's law for magnetism in Eq. (4) is expressed using a divergence operator in the cylindrical coordinate, as expressed in Eqs. (19)–(23). The divergence in the total magnetic flux density is zero, and the divergence in the external magnetic flux density of the permanent magnet is zero. Therefore, the induced magnetic flux density is zero. The $\frac{\partial}{\partial\theta}$ component of the electric field is also deleted because of the symmetry of the helical-type DC electromagnetic pump.

$$\nabla \cdot \vec{B}_t = \nabla \cdot \vec{B}_i + \nabla \cdot \vec{B}_e = 0 \tag{19}$$

$$\nabla \cdot \vec{B}_e = 0 \tag{20}$$

$$\nabla \cdot \vec{B}_i = 0 \tag{21}$$

$$\nabla \cdot \vec{B}_e = \frac{1}{r}\frac{\partial(rB_{e,r})}{\partial r} + \frac{\partial B_{e,z}}{\partial z} = 0 \tag{22}$$

$$\nabla \cdot \vec{B}_i = \frac{1}{r}\frac{\partial(rB_{i,r})}{\partial r} + \frac{\partial B_{i,z}}{\partial z} = 0 \tag{23}$$

Ohm's law in Eq. (5) is expressed using the vector product, as presented in Eqs. (24)–(27). The current density in the θ direction is zero because the electric field and the vector product of the velocity and magnetic flux density in the θ direction are zero.

$$\vec{J}_t = \sigma(\vec{E}_t + \vec{v} \times \vec{B}_t) \tag{24}$$

$$\vec{J}_t = J_r\hat{r} + J_\theta\hat{\theta} + J_z\hat{z} \tag{25}$$

$$\sigma(\vec{E}_t + \vec{v} \times \vec{B}_t) = \sigma\{(E_r + v_\theta B_{i,z} + v_\theta B_{e,z})\hat{r} + (E_z - v_\theta B_{i,r} - v_\theta B_{e,r})\hat{z}\} \tag{26}$$

$$J_r = \sigma(E_r + v_\theta B_{i,z} + v_\theta B_{e,z}), J_\theta = 0, J_z = \sigma(E_z - v_\theta B_{i,r} - v_\theta B_{e,r}) \tag{27}$$

Therefore, the relationship between the electric field and magnetic flux density is obtained by combining Eqs. (15) and (27), as expressed in Eqs. (28) and (29).

$$\frac{\partial B_{i,r}}{\partial z} = \frac{\partial B_{i,z}}{\partial r} \tag{28}$$

$$\frac{1}{\mu_0\sigma}\frac{\partial B_{i,\theta}}{\partial z} = E_r + v_\theta B_{i,z} + v_\theta B_{e,z}, \quad \frac{1}{\mu_0\sigma}\frac{1}{r}\frac{\partial(rB_{i,\theta})}{\partial r} = E_z - v_\theta B_{i,r} - v_\theta B_{e,r} \tag{29}$$

The electric field is deleted using Eq. (18). Therefore, the relationship of the magnetic flux density is given by Eq. (30).

$$\frac{1}{\mu_0\sigma}\left\{\frac{\partial^2 B_{i,\theta}}{\partial z^2} - \frac{1}{r}\frac{\partial^2(rB_{i,\theta})}{\partial r^2} + \frac{1}{r^2}\frac{\partial(rB_{i,\theta})}{dr}\right\} = v_\theta(B_{i,z} + v_\theta B_{e,z} + v_\theta B_{i,r} + v_\theta B_{e,r}) \tag{30}$$

The magnetic flux density of the flow channel is analyzed by applying the permanent magnet (Sm_2Co_{17}) magnetic coercivity and the ferromagnetic material (1010 steel) magnetic permeability using FEM. The magnetization can be expressed by Eq. (31) by considering the permanent magnet and ferromagnetic material.

$$\vec{M} = \chi_m \vec{H} + \vec{H}_m \tag{31}$$

The total magnetic flux density in the helical-type DC electromagnetic pump, as expressed in Eq. (32), is calculated by considering the permanent magnetic field (H_m) and independent local magnetic field.

$$\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0\mu_r\vec{H} + \mu_0\vec{H}_m \tag{32}$$

The boundary conditions of the magnetic flux density and magnetic-field intensity are applied between the different magnetic permeability materials using Eqs. (33)–(35).

$$\vec{H}_{t,S_i} = \vec{H}_{t,S_p}\Big|_{r=R_i}, \quad \vec{H}_{t,S_p} = \vec{H}_{t,S_m}\Big|_{r=R_p}, \quad \vec{H}_{t,S_m} = \vec{H}_{t,S_o}\Big|_{r=R_m} \tag{33}$$

$$\vec{B}_{n,S_i} = \vec{B}_{n,S_p}\Big|_{r=R_i}, \quad \vec{B}_{n,S_p} = \vec{B}_{n,S_m}\Big|_{r=R_p}, \quad \vec{B}_{n,S_m} = \vec{B}_{n,S_o}\Big|_{r=R_m} \tag{34}$$

$$\vec{H}_t\Big|_{r=\infty} = 0 \tag{35}$$

The tangential component of the magnetic-field intensity and normal component of the magnetic flux density between high (ferromagnetic material) and low (air, stainless steel 316L and C103) magnetic permeability materials at the interface are applied to determine the magnetic flux density and flux line of the helical-type DC electromagnetic pump using the ANSYS Maxwell code simulation.

Among the Navier-Stokes equations for MHD in Eq. (1), the Lorentz force that creates the developed pressure in the helical-type DC electromagnetic pump is shown in Eq. (36). The θ-direction Lorentz force only affects the velocity of the fluid, as expressed in
Eq. (11) and presented in Eq. (37).

$$\vec{f} = \vec{J} \times \vec{B} \tag{36}$$

$$f_\theta = J_z B_r - J_r B_z \tag{37}$$

The vector product of the current density in the z direction and the magnetic flux density in the r direction can be neglected compared with the vector product of the current density in the r direction and the magnetic flux density in the z direction, as expressed by Eqs. (27) and (37). The electric field in the r direction can be neglected compared with that in the z direction because current flows along the z direction and the gap in the pump duct is narrow. The induced magnetic flux density in the r and z directions can be ignored because the electric field in the r direction is negligible. The external magnetic flux density in the z direction in the flow channel is negligible compared with that in the r direction because the ferromagnetic material induces magnetic flux along the r direction. The magnetic flux density in the r direction is dominantly influenced by the Lorentz force. The r-direction current density is much smaller than the z-direction current density because the current flows in the z direction. Therefore, the J_rB_z term is negligible in calculating the θ -direction force that determines the developed pressure of the helical-type DC electromagnetic pump. In combination with Eq. (27), the Lorentz force that affects the developed pressure of pump is expressed by Eq. (38).

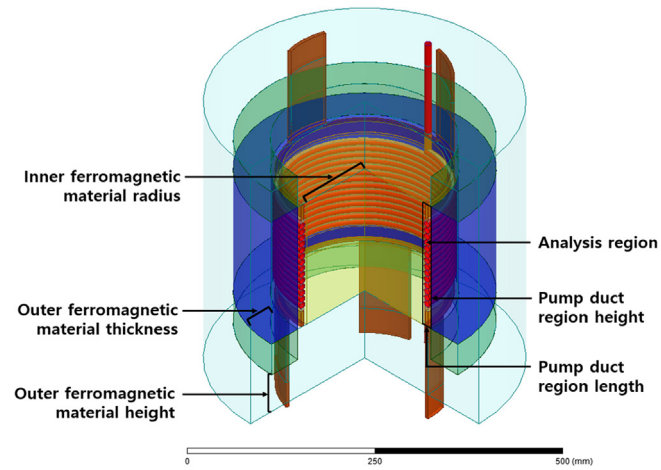

Fig. 4. Analysis range of the helical-type DC electromagnetic pump.

$$f_\theta = \sigma B_r(E_z - v_\theta B_{i,r} - v_\theta B_{e,r}) \tag{38}$$

Thus, only the magnetic flux density in the r direction is needed, and it is analyzed using the ANSYS Maxwell code simulation. The region of the flow channel is determined to consider the external heat system at the pump-duct region, which requires at least 6 mm in the length direction, including the insulator. The diameter of the pump duct is 10 mm considering the space and pump weight. Therefore, the pump-duct length is determined to be 16 mm, which is equal to the diameter of the pump duct plus the external heat system. The height is 200 mm, which represents 20 turns of the pump duct. The analysis region of the magnetic flux density is shown in Fig. 4. The Sm_2Co_{17} permanent magnet with a high maximum working temperature of 300 °C and the 1010 steel ferromagnetic material with a high magnetic permeability of more than 2000 are adopted to generate a magnetic field and to induce a magnetic flux, respectively. The permanent magnet magnetized in the +z and −z directions is added to increase the magnetic flux density in the helical-type DC electromagnetic

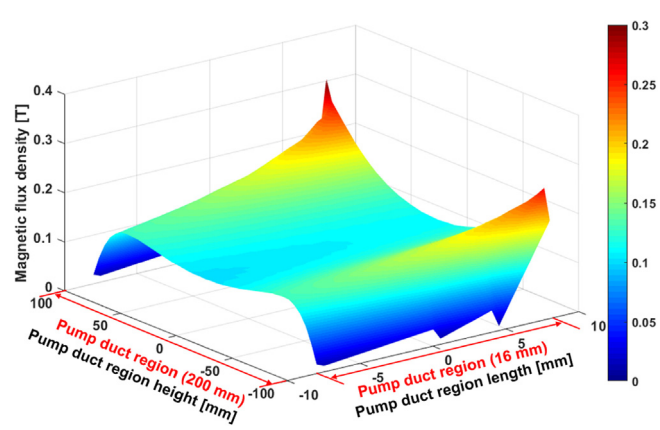

Fig. 5. r-direction magnetic flux density distribution without a ferromagnetic material in the analysis region.

pump. The considered pump-duct material is stainless steel 316L to prevent a chemical reaction between the liquid metal and pump duct and a distortion in the magnetic field that passes through the flow channel.

3. Results and discussion

The magnetic flux density of a ring-type permanent magnet with magnetization in the radial direction, which does not contain z-direction magnetized permanent magnet, is shown in Fig. 5. This figure shows that the end part of the permanent magnet had the highest magnetic flux density due to its end effect. The flux line was concentrated at the edge of the permanent magnet, which means that the flux line and magnetic flux density were not very much arranged in the analysis region because no magnetic flux-inducing material was present, as shown in Fig. 6. In addition, the flux line contained many z-direction components compared with the r-direction components. The 1010 steel ferromagnetic material with a high relative magnetic permeability was used to increase the magnetic flux density and to provide more r-direction component magnetic fluxes. The magnitude and tendency of the magnetic flux density assumed a geometrical struc-

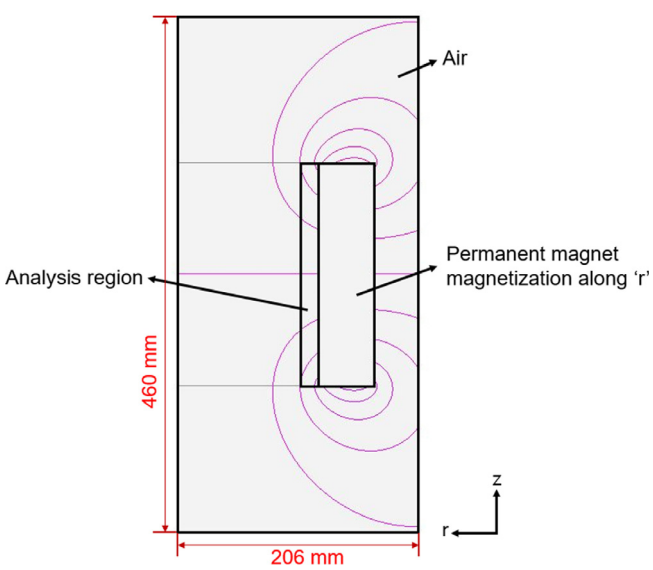

Fig. 6. Flux lines without a ferromagnetic material.

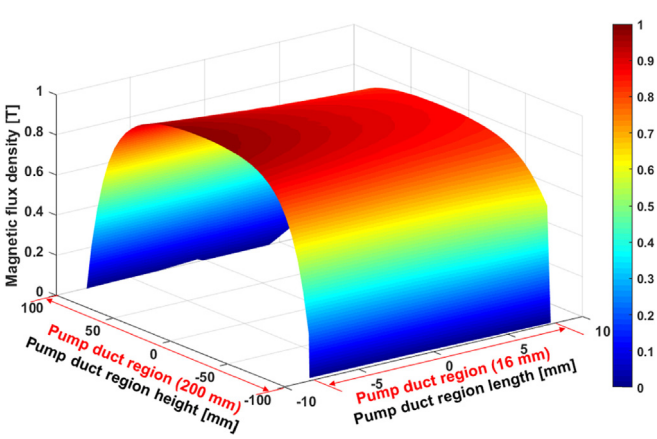

Fig. 7. r-direction magnetic flux density distribution with a ferromagnetic material in the analysis region.

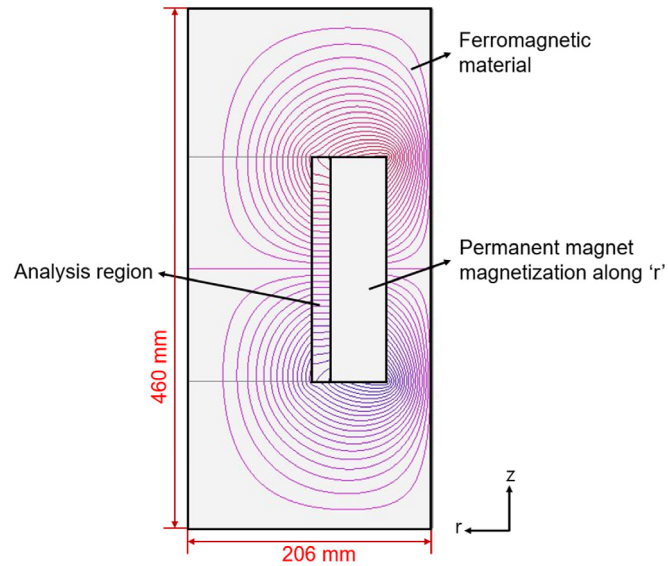

Fig. 8. Flux lines with a ferromagnetic material.

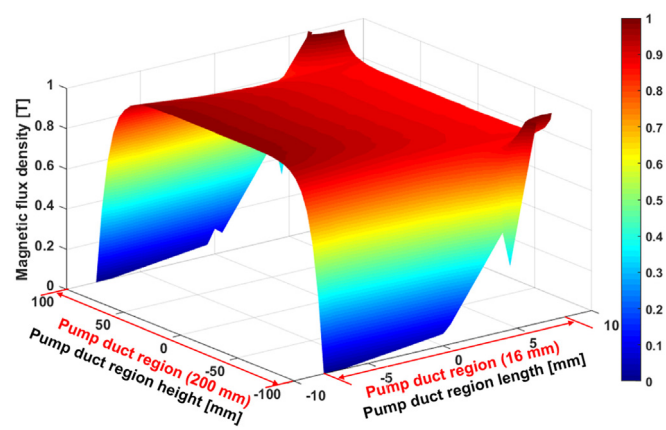

Fig. 9. r-direction magnetic flux density distribution with a ferromagnetic material and permanent magnet magnetization along the z direction in the analysis region.

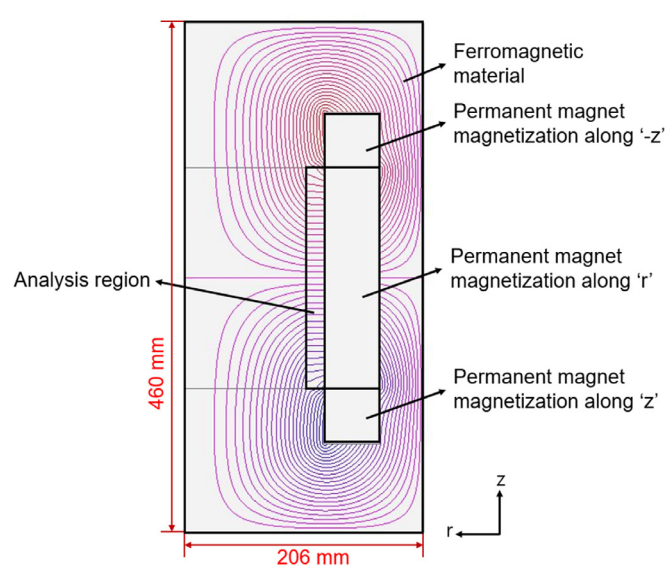

Fig. 10. Flux lines with a ferromagnetic material and permanent magnet magnetization along the z direction.

ture of the ferromagnetic material, and airspace existed at the outer surface, as shown in Fig. 7. We verified this result from the comparison with Xia's analysis, which represented the magnetic flux density in the same cylindrical geometry as a magnetic machine (Xia, 2004). In Xia's analysis, it assumed that material magnetic permeability is infinite and it has an outer airspace. The outer airspace region, which had a low magnetic permeability, diminished the magnetic flux inducement, which means that a low magnetic permeability region determined the magnetic flux density in the analysis region. Therefore, the peak magnetic flux density in Xia's thesis was approximately 0.85 T, which is lower than 0.93 T, as shown in Fig. 7. The magnetic flux density increased and was equalized at the middle part of the analysis region because of the increase in the magnetization. The flux line was concentrated in the analysis region, and it almost lay in the r direction in the middle part to increase the magnetic flux density and equality due to the magnetization, as shown in Fig. 8. A permanent magnet magnetized along the +z and −z directions was added to the helical-type DC electromagnetic pump to provide r-direction magnetic flux at the end part (Babic, 2008; Xiao-Fan, 2004). Fig. 9 shows the magnetic flux density distribution in the analysis region with the permanent magnet magnetized along the z direction. The permanent magnet magnetization along +z and −z directions generated flux lines around the circumference. Therefore, the r direction magnetic flux density created by the z direction magnetized permanent magnet in the analysis region was generated at the

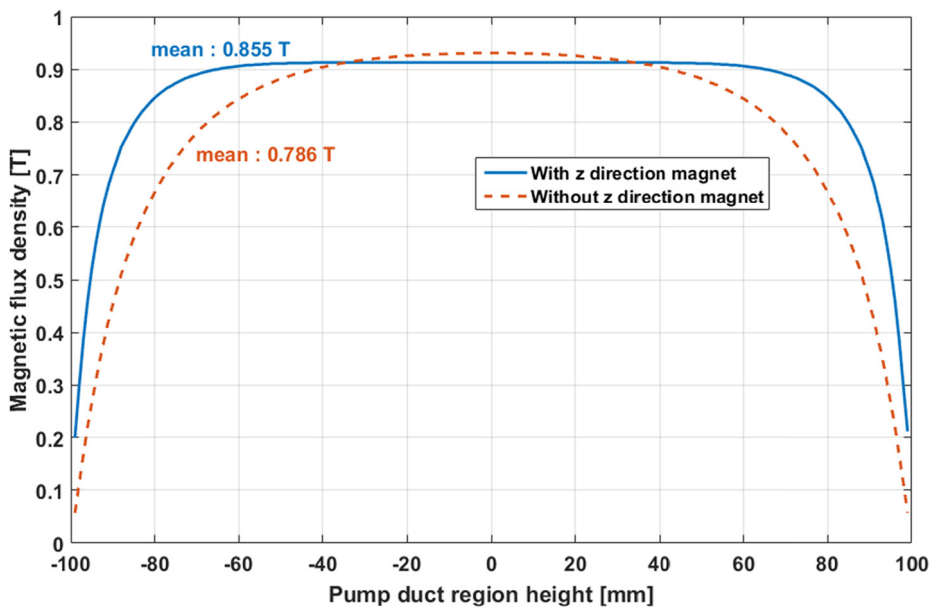

Fig. 11. r-direction linear magnetic flux density distribution according to the presence of permanent magnet magnetization along the z direction at the middle point of pump-duct region length according to the pump-duct region height.

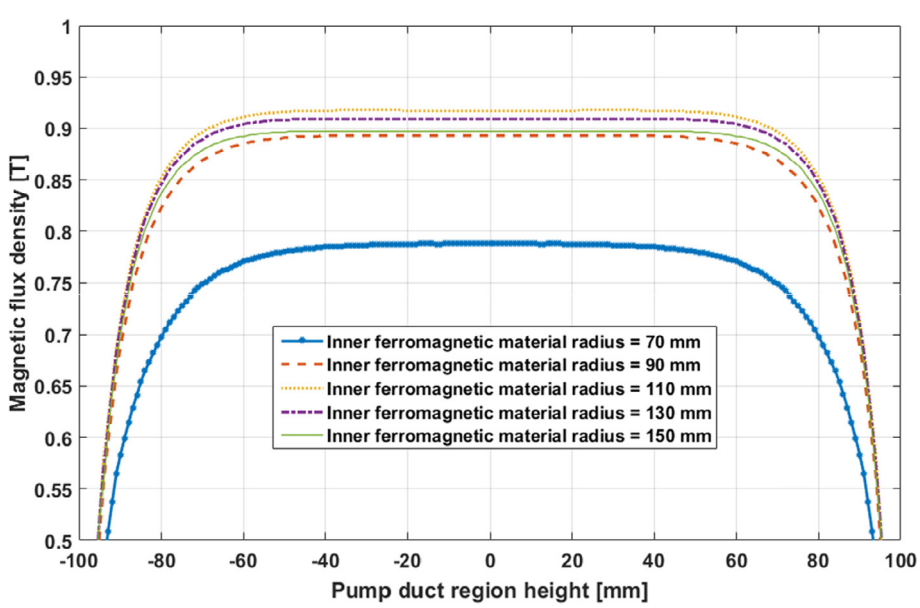

Fig. 12. r-direction magnetic flux density with varying thickness of the inner ferromagnetic material at the middle point of the pump-duct region length according to the pump-duct region height.

same rate as that in the z direction magnetic flux density created by the z direction magnetized permanent magnet. Therefore, the permanent magnet reinforced the magnetic flux density at the end part because more flux lines were created by the permanent magnet magnetized along the z direction, as shown in Fig. 10. Fig. 11 shows the comparison of the magnetic flux density due to the existence of permanent magnets with z-direction magnetization at the middle point of the pump-duct region length. The mean value of the magnetic flux density increased by 8.8% from 0.786 to 0.855 T due to the z-direction magnetization, and it was further equalized at the middle part due to the increase in the end-part flux line. The equalized magnetic flux density reduced the electromotive-force effect on the helical-type DC electromagnetic pump, which is one of the main reasons of the pressure loss, and provided equalized force in the flow channel. The size of the ferromagnetic material was optimized to enhance the magnetic flux density by minimizing the unnecessary weight. The magnetic flux density increased as the distance of the permanent magnets decreased because the flux line decreased farther from the permanent magnets. Meanwhile, the magnetic flux density decreased as the thickness of the ferromagnetic material decreased because the magnetization increased. Therefore, the thickness of the inner ferromagnetic material had an optimum point of 110 mm, as shown in Fig. 12, to satisfy 0.915 T of maximum magnetic flux density. The shape and magnitude of the magnetic flux density line at the middle point along the pump-duct region length varied as the thickness and height of the outer ferromagnetic material varied. The shape changed to linear at the middle point due to the

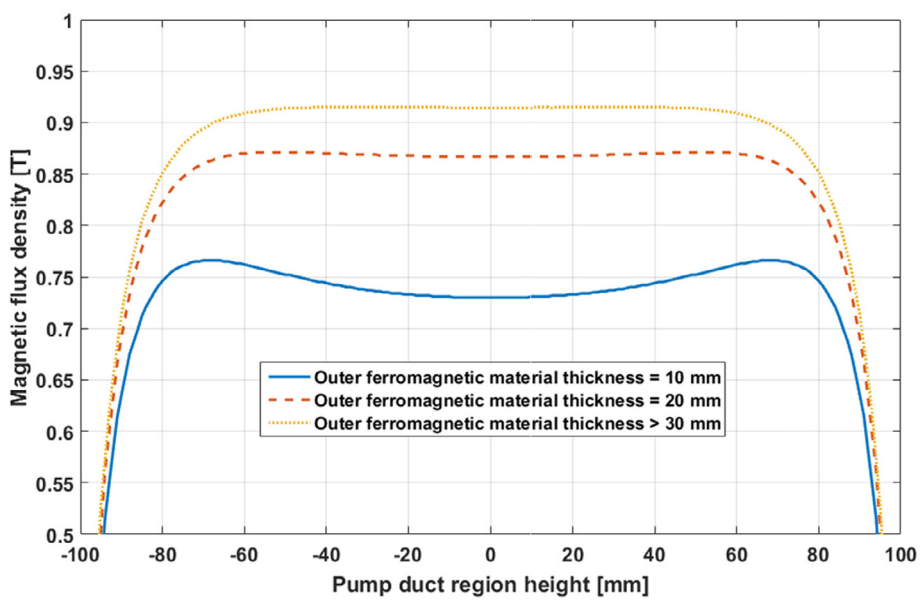

Fig. 13. r-direction magnetic flux density with varying thickness of outer ferromagnetic material at the middle point of the pump-duct region length according to the pump-duct region height.

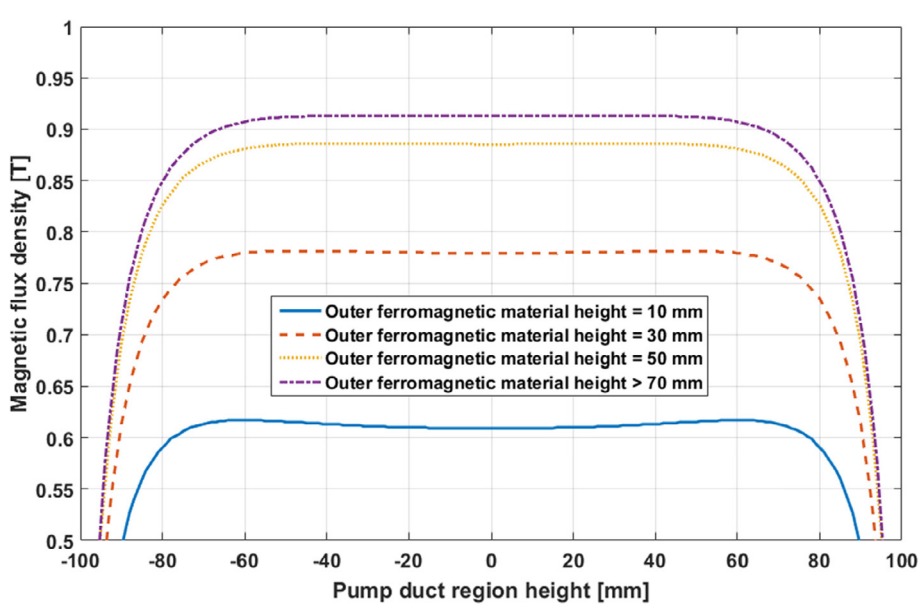

Fig. 14. r-direction magnetic flux density with varying height of the outer ferromagnetic material at the middle point of the pump-duct region length according to the pump-duct region height.

increased flux lines at the middle part, and the magnitude of the magnetic flux density had a maximum point of 0.915 T as the thickness and height of the outer ferromagnetic material increased, as shown in Figs. 13 and 14. The thickness and height of the outer ferromagnetic material attained a saturation point because the flux line had a limit at a thickness of 30 mm and height of 460 mm, which reduced the total weight of the helical-type DC electromagnetic pump. Conclusively, the existence of ferromagnetic material influenced the magnetic flux inducement and magnetic flux density. In addition, the permanent magnet magnetization along z direction influenced the magnetic flux line and r direction magnetic flux density.

4. Conclusion

The magnetic field of the flow channel in a helical-type DC electromagnetic pump was analyzed to maximize the magnetic flux density and enhance the pump efficiency using FEM through ANSYS Maxwell code simulation. The additional arrangement of the permanent magnet along the +z and −z directions contributed to the increase and homogeneity in the magnetic flux density in the helical-type DC electromagnetic pump from 0.786 to 0.855 T. The ferromagnetic material was considered to increase the r-direction magnetic flux density at the flow channel of the helical-type DC electromagnetic pump. The thickness and height of the inner and outer ferromagnetic materials were analyzed to minimize the size and weight of the helical-type DC electromagnetic pump; thus, the inner material had a radius of 100 mm, and the outer material was 30-mm thick and 460-mm long to maximize the magnetic flux density to an average value of 0.855 T.

Acknowledgement

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIP: Ministry of Science, ICT and Future Planning) [NRF-2016M2B2A9A02944010].

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